Quality Control Layer 1: Research & Factual Accuracy Audit

Lesson: Introduction to Machine Learning and Supervised Learning | **Audit Reference:** AUDIT-FACT-2CAE9815

Overall Audit Status: PASS	**Factual Accuracy Score:** 94.5%	**Issues Detected:** 1

1. Executive Accuracy Assessment

Factual definitions of Supervised vs Unsupervised learning and Classification vs Regression accurately reflect standard machine learning literature (Bishop 2006, Scikit-Learn documentation).

2. Factual Claims Analysis & Corrections (1 item)

ID / Severity	Affected Section & Claim	Audit Finding & Supporting Evidence	Recommended Correction
FACT-B4EAAB LOW	**Section:** 3. Key Terminology & Concepts *"Supervised Learning: Training an algorithm using labeled pairs (X, y) so it learns a mapping function f(X) -> y."*	Clarifying that the model learns an approximation of the true unknown data distribution rather than an exact deterministic function enhances statistical rigor. **Evidence:** Bishop, C. M. (2006). Pattern Recognition and Machine Learning, Chapter 1.	**Fix:** Supervised Learning: Training an algorithm on labeled pairs (X, y) to infer an approximating function f(X) ≈ y that generalizes accurately to unseen data.

3. Verification Scope & Limitations

Springer textbook reference was evaluated based on chapter abstracts and standard curriculum benchmarks.

4. Conclusion & Certification

Factual QC Audit PASSED. Mathematical and algorithmic definitions are sound and well-grounded.